AE 481 Team 01 Assignment 7

November 3, 2025

1 Design Adjustments

Based on the feedback from the preliminary design review presentation, the team will be making the following adjustments:

1. The team plans on revising the fuselage sizing, especially the width, to ensure the fuselage volume can comfortably accommodate all fuel tanks, internal ordnance, and any other internal components. This was partially due to comments on the compactness of the F/A-67 internal weapons bay, and the orientation of ordnance. The revised weight estimate will change the fuel tank sizing, which can introduce more space for other components like the duct.
2. The airfoil optimization will be redone to be more representative of the actual flight conditions. This includes accounting for wing sweep. Currently, the airfoil optimization is representative of an unswept wing, which is different than the swept operating condition of the actual F/A-67. Ideally, this improves airfoil manufacturability by increasing leading edge radius. Additionally, based on feedback on inconsistencies on the airfoil drag polar, the CFD solution will be refined to produce better converged results.
3. The aircraft static margin will be increased based on feedback on the negative static stability of the preliminary design. This will make it simpler to justify the controllability of the aircraft, make the aircraft less tail heavy, and improve landing gear angles such as tip-over. This largely relies on the component weight buildup for a refined CG estimate.

2 Mission Fuel Refinement

The weight fractions for cruise and climb mission segments were refined in this assignment using a multi-segment approach.

2.1 Cruise Weight Fraction

Cruise mission segments are modeled as a cruise climb over a fixed distance. The distance traveled is subdivided into n sub-segments, with each assume to be flown at a constant altitude. The entire cruise is flown at a fixed Mach number of 0.84. The range traveled during a cruise subsegment is derived as follows.

$$R = \int_{W_{i+1}}^{W_i} \frac{V}{c} \frac{L}{D} \frac{dW}{W}$$

Velocity is substituted and L/D is replaced by their respective coefficients.

$$R = \int_{W_{i+1}}^{W_i} \frac{1}{c}\sqrt{\frac{2W}{\rho S C_L}}\frac{C_L}{C_d}\frac{dW}{W}$$

Integrating yields equation (1). The equation can be rearranged to find the subsegment weight fraction W_{i+1}/W_i.

$$R = \frac{2}{c}\sqrt{\frac{2}{\rho S}}\frac{C_L^{1/2}}{C_D}\left(\sqrt{W_i} - \sqrt{W_{i+1}}\right) \tag{1}$$

Since the cruise segment is divided into sub-segments, the range R, thrust-specific fuel consumption c, and wing reference area S, are known. The weight at the start of the cruise segment W_i, or sub-segment is also known. To reduce fuel burn during a sub-segment, $\left(\sqrt{W_i} - \sqrt{W_{i+1}}\right)$ is minimized. This can be done by maximizing all other unknowns on the right hand side of equation (1). For a given W_i, the altitude that maximizes the expression in equation (2) is found using MATLAB's `fmincon`. This is determined at a fixed cruise Mach number of 0.84, and for a given aircraft (i.e., for some S, C_{D0}, etc.).

$$\frac{1}{\rho S}\frac{C_L^{1/2}}{C_D} \tag{2}$$

The assumption is made that the aircraft's velocity does not change significantly during the cruise sub-segment, and that the change in altitude during the sub-segment is small. This holds as long as the change in weight is small. The weight fraction for the entire cruise segment is the product of the weight fractions for all sub-segments as shown in equation (3)

$$\text{Weight Fraction} = \prod_{i=1}^{n}\frac{W_{i+1}}{W_i} \tag{3}$$

2.2 Climb Weight Fraction

Climb mission segments follow a similar sub-segment procedure as the cruise mission segment. Instead of subdividing a given range, altitude is subdivided into n sub-segments. Sub-segment weight fraction is determine using the energy method.

$$\frac{W_{i+1}}{W_i} = \exp\frac{-c\Delta h_e}{V(1 - D/T)} \tag{4}$$

h_e in equation (4) is the energy height which is defined as

$$h_e = h + \frac{V^2}{2g} \tag{5}$$

The velocity flown the climb segment is the velocity for the best rate-of-climb which is found using equation (6) from Raymer [1]. Lift is assumed to be equal to weight for small climb angles, which can be used to calculate drag. Wing loading is calculated for the beginning of a sub-segment. The thrust-to-weight ratio and the thrust must be determined numerically since the aircraft's velocity influences thrust lapse, which changes thrust-to-weight, which changes the velocity for best rate-of-climb. The change in velocity as the altitude changes is assumed to be small.

$$V = \sqrt{\frac{W/S}{3\rho C_{D0}}\left(\frac{T}{W} + \sqrt{\left(\frac{T}{W}\right)^2 + 12C_{D0}K}\right)} \tag{6}$$

The weight fraction for the entire climb segment is determined using equation (3). Credit is also taken for the distance traveled during the climb. This is done by calculating the specific excess power using equation (7).

$$P_s = \frac{TV - DV}{W_i} \tag{7}$$

Dividing the change in energy height by the specific excess power yields the time taken to fly a sub-segment. Assuming a small flight-path angle, multiplying by V yields the distance traveled for a sub-segment as shown in equation (8).

$$\Delta x = \frac{\Delta h_e}{P_s} V \tag{8}$$

An additional consideration is the determination of the climb height. The climb should transition to a cruise segment at the altitude where the aircraft weight at the end of the climb yields an identical altitude to begin an optimal cruise.

2.3 Future Refinements

The refined climb weight fraction method yields velocities for best rate of climb that are transonic or even supersonic. This is attributed to the current drag polar model, which assumes a fixed C_{D0} and K. Future refinements will include both a component drag buildup for subsonic and supersonic C_{D0}, and model for transonic drag rise and wave drag. This should yield more realistic velocities for best rate-of-climb.

3 Component Weight and CG Estimation

A detailed weight component buildup is done using equations from Raymer. CG estimations of these components are also estimated, with smaller items falling into an "all else" category, from Raymer. These metrics are then used to update our empty weight and center of gravity predictions. The following sections detail the various components of the aircraft weight. All fudge factors are from Table 15.4 in Raymer, and the highest of each is assumed for a more conservative estimate.

3.1 Airframe Weight

This section will discuss the calculations needed for the weight of various parts of the airframe.

3.1.1 Wing Weight

The wing weight is found using Eq 9:

$$W_{\text{wing}} = K_{\text{comp}} * 0.0103 K_{dw} K_{vs} (W_{dg} N_z)^{0.5} S_w^{0.622} AR^{0.785} (t/c)_{\text{root}}^{-0.4} (1+\lambda)^{0.05} cos(\Lambda_{c/4})^{-1} S_{csw}^{0.04} \tag{9}$$

where K_{comp} is a composite fudge factor, K_{dw} is a delta wing factor, K_{vs} is a variable sweep factor, W_{dg} is the flight design gross weight, N_z is the ultimate load factor, given by 1.5 times the limit load factor, S_w is the trapezoidal area of the wing, AR is the aspect ratio, $(t/c)_{\text{root}}$ is the thickness to chord ratio at the root of the wing, λ is the taper ratio, $\Lambda_{c/4}$ is the sweep at quarter-chord, and S_{csw} is the wing control surface area.

The F/A-67 is not a delta wing and does not have variable sweep, so those constant factors are set to 1. As assumed in all equations, the flight design gross weight is assumed to be the empty weight of the aircraft plus 40% of the fuel weight. The limit load factor is 8. The rest of the parameters are determine using wing geometry.

3.1.2 Fuselage

The fuselage weight is found using Eq 10

$$W_{\text{fuselage}} = K_{\text{carrier}} K_{\text{comp}} 0.499 K_{dwf} W_{dg}^{0.35} N_z^{0.25} F_{\text{L}}^{0.5} F_{\text{D}}^{0.849} F_{\text{W}}^{0.685} \tag{10}$$

where K_{carrier} is a fudge factor applied for carrier aircraft, K_{dwf} is a constant for delta wing aircraft (assumed to be 1), and F_{L}, F_{D}, F_{W}, are the structural fuselage length, depth, and width respectively.

3.1.3 Horizontal Tail

The horizontal tail weight is found using Eq 11

$$W_{\text{HT}} = K_{\text{comp}} 3.316 (1 + F_w/B_{\text{HT}})^{-2} \frac{W_{dg} N_z}{1000}^{0.260} S_{\text{HT}}^{0.806} \tag{11}$$

where F_w is the width of the fuselage at the horizontal tail, B_{HT} is the span of the tail, and S_{HT} is the area of the horizontal tail.

3.1.4 Vertical Tail

The vertical tail weight is found using Eq 12

$$\begin{aligned} W_{\text{VT}} = 2K_{\text{comp}} 0.452 K_{rht} (1 + H_t/H_v)^{0.5} (W_{dg} N_z)^{0.488} S_{\text{VT}}^{0.718} M^{0.341} L_t^{-1.0} \\ \times (1 + S_r/S_{\text{VT}})^{0.348} AR_{\text{VT}}^{0.223} (1 + \lambda_{\text{VT}})^{0.25} cos(\Lambda_{\text{VT}})^{-0.323} \end{aligned} \tag{12}$$

Where K_{rht} is a constant for a rolling horizontal tail (1.047 given by Raymer), H_t/H_v is the horizontal tail height above the fuselage divided by the vertical tail height resulting in a zero term. The rest of the terms follow the same conventions as the wing equation, with the subscript VT denoting vertical tail. The equation is multiplied by two to consider both vertical tails on the aircraft.

3.2 Engine Related Components

This section discusses the engine related components of the weight buildup, as given by Raymer. Engine selection accounts for engine offtakes, as discussed in previous assignments.

3.2.1 Engine Mounts

The engine mounts weight is calculated using Eq.13:

$$W_{\text{Engine Mounts}} = K_{\text{comp}} 0.013 N_{\text{en}}^{0.795} (N_{\text{en}} * T_{\text{e}})^{0.579} N_z; \tag{13}$$

where N_{en} is the number of engines and T_{e} is the thrust per engine in lb.

3.2.2 Engine Section

The weight of the engine sections are determined using Eq 14:

$$W_{\text{Engine Section}} = K_{\text{comp}} 0.01 W_{\text{en}}^{0.717} N_{\text{en}} N_z \tag{14}$$

where W_{en} is the weight of one engine.

3.2.3 Air Induction

The engine air induction weight is calculated using Eq. 15:

$$W_{\text{Air Induction}} = K_{\text{comp}} 13.29 K_{vg} L_d^{0.643} K_d^{0.182} N_{\text{en}}^{1.498} L_s/L_d{}^{-0.373} D_{\text{e}} \tag{15}$$

where K_{vg} is a constant for variable geometry, L_d is the air induction duct length, L_s is the length of the portion of the duct that is not split, K_d is a duct geometry constant, and D_{e} is the diameter of the engine.

K_{vg} is assumed to be one since the duct does not vary in geometry. L_s is assumed to be equal to L_d since the ducts never combine, there is only one duct for each engine. K_d is assumed to be 2.75 from Figure 15.3 in Raymer based on our inlet geometry.

3.2.4 Engine and Oil Cooling

Eq. 16 and Eq. 17 calculate engine and oil cooling weights respectively.

$$W_{\text{Engine Cooling}} = 4.55 D_{\text{e}} L_{sh} N_{\text{en}} \tag{16}$$

$$W_{\text{Oil Cooling}} = 37.82 * N_{\text{en}}^{1.023} \tag{17}$$

were L_{sh} is the length of the engine cooling shroud.

3.2.5 Engine Controls and Starter(Pneumatic)

The final components of the engines, beyond the engines themselves, are given by Eq. **??** and 19:

$$W_{\text{Engine Controls}} = 10.5 N_{\text{en}}^{1.008} L_{\text{ec}}^{0.222} \tag{18}$$

$$W_{\text{Starter(pneumatic)}} = 0.025 T_{\text{e}}^{0.760} N_{\text{en}}^{0.72} \tag{19}$$

where L_{ec} is the routing distance from engine front to cockpit, total distance if multi-engine.

3.3 Equipment

This section discusses the weight of equipment in the aircraft.

3.3.1 Flight Controls

The weight of the flight controls is calculated using Eq 20:

$$W_{\text{Flight Controls}} = 36.28 M^{0.003} S_{\text{cs}}^{0.489} N_s^{0.484} N_{\text{c}}^{0.127} \tag{20}$$

where S_{cs} is the total control surface area, N_s is the number of flight control systems, N_{c} is the number of crew.

3.3.2 Instruments

The weight of the instruments is calculated using Eq 21:

$$W_{\text{Instruments}} = 8.0 + 36.37 N_{\text{en}}^{0.676} N_t^{0.237} + 26.4(1 + N_{\text{ci}})^{1.356} \tag{21}$$

where N_t is the number of fuel tanks, currently 5 per the CAD, and N_{ci} is the crew equivalence number which is one for a single crew member.

3.3.3 Hydraulics

The hydraulics weight is calculated using Eq 22:

$$W_{\text{Hydraulics}} = 37.23 * K_{vsh} * N_u^(0.664); \tag{22}$$

where K_{vsh} is a variable wing sweep constant assumed to be one and N_u is the number of hyrdualic utility functions. Raymer estimates N_u to be between 5 and 15, given the F/A-67, and N_u of 15 was chosen.

3.3.4 Electrical

The weight of the electrical system is calculated using Eq. 23:

$$W_{\text{Electrical}} = 172.2 K_{mc} R_{kva}^{0.152} * N_{\text{c}}^{0.10} L_a^{0.10} N_{\text{gen}}^{0.091} \tag{23}$$

where K_{mc} is an electrical failure constant, R_{kva} is the system electrical rating, L_a is the electrical routing distance from generators to avionics to the cockpit, and N_{gen} is the number of generators.

The electrical failure constant is assumed to be 1.45, as given by Raymer. Also given by Raymer, the system electrical rating for fighter jets between 110-160 KvA, so it is assumed to be 160 KvA since a higher system electrical rating increases the weight of the system for a more conservative estimate. The number of generators is typically assumed to be the same as the number of engines.

3.3.5 Furnishings

The weight of the furnishings, including crew seats, is only dependent on the number of crew and is given by Eq. 24:

$$W_{\text{Furnishings}} = 217.6N_{\text{c}} \tag{24}$$

3.3.6 Air Conditioning

The air conditioning system is dependent on the uninstalled weight of avionics. The requirement of 2500 lb from the RFP is taken to be the installed avionics weight. The uninstalled avionics weight is calculated using Eq. 25 and the air conditioning weight is calculated using Eq. 26.

$$W_{\text{Uninstalled Avionics}} = {\frac{W_{\text{Avionics}}}{2.117}}^{1/0.933} \tag{25}$$

$$W_{\text{Air Conditioning}} = 201.6 {\frac{W_{\text{Uninstalled Avionics}} + 200N_{\text{c}}}{1000}}^{0.735} \tag{26}$$

3.3.7 Handling Gear

The handling gear is given by Eq. 27:

$$W_{\text{Handling Gear}} = 3.2 * 10^{-4} W_{dg} \tag{27}$$

3.4 Fuel System

The weight of the entire fuel system is calculated using Eq. 28:

$$W_{\text{Fuel System}} = 7.45V_t^{0.47}(1 + V_i/V_t)^{-0.095}(1 + V_p/V_t)N_t^{0.066}N_{\text{en}}^{0.052}(T \times TSFC/1000)^{0.249} \tag{28}$$

where V_t is the total volume of the tanks, V_i is the volume of integral tanks, V_p is the volume of self-sealing tanks, T is the maximum thrust of the combined engines, and $TSFC$ is the thrust specific fuel consumption as maximum thrust. All tanks are assumed to be self-sealing for better fuel tank protection.

3.5 Avionics Weights

The avionics weight is fixed at 2500 lb. The primary avionics systems weights are detailed in Table 1. The JPALS system uses an onboard GPS to interface with the JPALS network. The H-764 is the GPS chosen as it has been proven on most military aircraft [2]. The F/A-18 Advanced Mission Computer from General Dynamics does not have a publicly available weight, but the JARVIS Mission Computer provides a good estimate [3]. There are four flight control computers as part of the quadruple redundant fly-by-wire system [4]. The rest of the avionics will be composed of other essential components. However, due to lack of information, the weights of the other components part of the 2500 lb avionics weight were unable to be found.

Table 1: Available Avionics Weights

	Weight of Avionics (lb)
AN/APG-81 Radar [5]	485.0
H-764 JPALS GPS	19.9
Network Centric Flight Control Computers	140.0
AN/ALQ-214 IDECM [6]	162.2
JARVIS Mission Computer	42.1
Total	849.2

3.6 Weight Component Buildup and CG Results

Table 2 details the results of the weight component buildup and their C.G. locations. Several weight components with no C.G. locations are lumped into an all-else category, as done by Raymer. Individual avionics components are placed within the avionics bay, with only the radar in a separate location. As such, the avionics without the radar and the all-else category are given a 'group' C.G. until more analysis can be done.

Table 2: Weight Component Buildup and C.G. Locations

Component	Weight (lb)	x C.G. Location (ft from the nose)	y C.G. Location (ft from the centerline)	z C.G. Location (ft from the ground)
Wings	6,894	31.76	0	7.10
Fuselage	5,690	19.73	0	7.10
Vertical Tails	2,143	43.90	0	12.98
Horizontal Tail	531	45.41	0	7.10
Nose Gear	840	11.20	0	1.61
Main Landing Gear	1,613	32.08	0	1.70
Installed Engines	8,942	37.92	0	7.10
All-else Empty	8,920	19.73	0	7.10
Air to Air Payload	2,460	25.06	0	4.66
Strike Payload	4,424	24.87	0	4.66

Installed engines weights are shown in Table 3. The all-else empty category is broken down in Table 4. The C.G. of individual components in these sections are not estimated.

Table 3: Installed Engines Components

Component	Weight (lb)
Engines	7,980
Engine Mounts	150
Engine Section	87
Air Induction	996
Cooling Systems	621
Starter (Pneumatic)	103

4 Refined Weight Estimation Results

Replacing the original Raymer empty weight fraction regression with the component weight estimate, and using the refined cruise and climb segments yields lower weights when compared to preliminary sizing results as seen in Table 5. First, the aircraft empty weight has reduced by roughly 6,000 lb. Second, the fuel required for both the air-to-air and strike missions have decreased by nearly a third. The reduction in empty weight

Table 4: All-Else Component Weights

Component	Weight (lb)
Avionics	2,500
Flight Controls	1,318
Engine Controls	43
Instruments	168
Hydraulics	225
Electrical	793
Furnishings	218
Air Conditioning	355
Handling Gear	17
Fuel Systems	2,287

comes from the implementation of the component weight estimate. Takeoff weight and fuel weight are both inputs to the empty weight estimate, but the empty weight appears to be largely insensitive to changes in takeoff weight and fuel weight. For example, increasing the takeoff weight to the preliminary value for the air-to-air mission only increases empty weight by roughly 200 lb. Therefore, the weight is largely dictated by other parameters such as the aircraft geometry. The reduction in fuel weight suggests that the new refined cruise and climb segments results in more efficient mission segments. The current combination of reduced fuel burn and a lower, fixed empty weight yields the lower, refined takeoff weights.

When using the Raymer empty weight fraction regression, increasing mission performance parameters like range results in increased fuel burn which yields higher takeoff weights and higher empty weight. However, since the component weight estimate for empty weight is relatively constant for a given aircraft, only fuel weight increases for improved mission performance. This results in lower takeoff weights than would be predicted using the preliminary regressions. Consequently, the design space needs to be reanalyzed for increased mission performance. This includes revisiting W/S-T/W sizing plots for feasibility.

Table 5: Refined Weights Comparison

	Refined Air-to-Air (lb)	Preliminary Air-to-Air (lb)	Refined Strike (lb)	Preliminary Strike (lb)
Empty	35,211	41,432	35,219	41,397
Fuel	23,693	32,274	21,902	30,270
Payload	2,460	2,460	4,424	4,424
Crew	200	200	200	200
Total	61,564	76,366	61,745	76,292

The empty weight will continue to be revised, including accounting for the effect of empty external fuel tanks.

5 Landing Gear Sizing

The F/A-67 features a tricycle landing gear configuration. The main landing gear has one tire, while the nose gear has twin wheels in order for the catapult to attach. This landing gear configuration was selected as it is the same as many historical carrier-based fighters, such as the F/A-18E/F [7].

5.1 Tire Sizing

Main landing gear tire size was calculated using equations 29 and 30 from Raymer, which take in the static weight on wheel and output the tire diameter and width. These formulas were used to size the main tires.

$$D_{\text{main}}\ (\text{in}) = 1.59\, W_W^{0.302} \tag{29}$$

$$W_{\text{main}}\ (\text{in}) = 0.0980\, W_W^{0.467} \tag{30}$$

Additional factors of safety of 7% and 25% are added to the weight on wheel, based on recommendations in Raymer and to account for future mass growth. This results in a main tire diameter of 3.12 ft, and a main tire width of 1.08 ft.

When using equations 29 and 30 for the nose tires, the diameter and width are smaller than historically recommended due to low nose wheel loads. Raymer states that the nose tire can be sized to between 60-100% of main wheel dimensions. Therefore, it is assumed that nose tire dimensions are 80% of main tire dimensions.

This results in a nose tire diameter of 2.50 ft, and a nose tire width of 0.87 ft. However, because the nose tires are split, the actual nose tire width is 0.43 ft.

Tire size was also calculated using equation 31, which calculates the kinetic energy dissipated on landing. However, this was not as constraining as the calculation using equations 29 and 30. These equations are valid for our aircraft, as it must conduct landings on non-carrier runways.

$$KE_{\text{braking}} = \frac{1}{2}\frac{W_{\text{landing}}}{g}V_{\text{stall}}^2 \tag{31}$$

5.2 Landing Gear Placement

The ratio of distances of the main and nose landing gear from the center of gravity was calculated using a moment balance. The distances were selected so that 80% of the static load is transmitted to the main landing gear.

The distance between the main gear and nose gear was manually selected to satisfy tipback and rotation angle constraints. Furthermore, the width of the main gear was selected to satisfy the overturn angle constraint. These angle constraints come from Raymer. Table 6 summarizes the F/A-67 landing gear angles and historical recommendations.

The wingstrike angle was selected to ensure there is no risk of wing strike on takeoff and landing. The angle constraint for wingstrike is from Roskam[8]. This shows that overturn is more constraining than wingstrike.

Table 6: Landing Gear Angles

	Angle (deg)	Historical (deg)
Tipback	20.16	>15
Main wheel to CG	25.1	>Tip-back
Overturn	54.96	<54
Wingstrike	15.8	>5

All constraints are satisfied except for the overturn angle, which is about 1 deg higher than the Raymer recommendation for carrier-based aircraft. This will be addressed by making the landing gear wider for future iterations.

5.3 Strut Height

The stroke height is calculated using equation 32 from Raymer. The resulting stroke height is 1.24 ft.

$$S = \frac{V_{\text{vertical}}^2}{2g\eta N_{\text{gear}}} - \frac{\eta_T}{\eta} S_T \tag{32}$$

Raymer lists that the oleo total length is approximately 2.5 times the stroke length. Therefore, the minimum strut length is 3.12 ft. The minimum length was selected in order to satisfy the RFP height constraint.

5.4 Stowed Gear

The landing gear are stowed to reduce drag in flight, as Roskam recommends stowing gear if the aircraft cruise speed is above 150 kts [8]. The main gear folds in half and is stowed inside the fuselage. The nose gear folds forward and up into the nose. Further investigation is required to ensure the gear properly fit within the fuselage.

6 Group Work Split

Table 7: Work Split

Person	Work Percentage	Brief Description of Work Done
Michael Chen	18%	Derived and implemented equations for improved fuel fractions. Implemented component weight buildup into sizing code.
Charles Choi	18%	Landing gear sizing and report writing.
Cristina Erskine	18%	Component weight buildup, GFE weights, and report writing.
James Gold	18%	Component weight buildup, GFE weights, assisted with energy methods and landing gear constraints, sizing, and report writing.
Santiago Ramos-Assam	18%	Landing gear sizing and report writing.
Thomas Sheridan	10%	Assisted with refined fuel fraction and report.

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